

## PS1, Part M.1: Discrete-State Formulation and Euler Equation

ECONS 509    Worked derivation for Question 2

**1. Household problem and fixed prices.** Let  $k$  denote current assets,  $k'$  next-period assets, and  $s$  the labor-efficiency state. The household solves

$$v(k, s) = \max_{k' \in \mathcal{K}, \bar{k} \geq k, c > 0} \left\{ u(c) + \beta \sum_{s'=1}^2 \mathcal{P}_{ss'} v(k', s') \right\}, \quad c = (1 + r^0)k + w^0 \bar{\epsilon}_s \bar{l} - k', \quad (1)$$

where  $u(c) = c^{1-\sigma}/(1-\sigma)$  and  $\bar{k} = 0$ . Use  $\beta = 0.96$ ,  $\sigma = 1.5$ ,  $\bar{l} = 1$ ,  $\bar{\epsilon} = (0.8, 1.2)$ , and  $\mathcal{P}_{ss'} = 0.5$  for every pair of shocks. Rows of  $\mathcal{P}$  condition on the current shock. For  $Y = zK^\alpha L^{1-\alpha}$ , take  $z = 1$ ,  $\alpha = 0.36$ ,  $\delta = 0.1$ , and  $K^0 = 5$ . The shock probabilities are  $(0.5, 0.5)$ , so  $L^* = \sum_s \pi_s^{shock} \bar{\epsilon}_s \bar{l} = 1$  and

$$w^0 = (1 - \alpha)z(K^0/L^*)^\alpha \simeq 1.1423762763, \quad r^0 = \alpha z(K^0/L^*)^{\alpha-1} - \delta \simeq 0.0285173311.$$

These prices stay fixed; mean household assets need not equal  $K^0$ .

**2. Vector form.** Let  $\mathcal{K} = \{k_1, \dots, k_N\}$ ,  $S = 2$ ,  $M = NS$ , and  $m(n, s) = (s-1)N + n$ . Stack the columns of  $\mathcal{V}$  in  $x = \mathcal{V}(\cdot)$ : all asset nodes in shock 1 precede those in shock 2.

| Object                   | Definition                                   | Dimension         |
|--------------------------|----------------------------------------------|-------------------|
| $\mathcal{V}$            | $\mathcal{V}_{n,s} = v(k_n, s)$              | $N \times S$      |
| $x = \mathcal{V}(\cdot)$ | Column-stacked values                        | $M \times 1$      |
| $\mathcal{R}$            | Current-state and savings-choice utility     | $M \times N$      |
| $\mathcal{G}$            | Maximizing savings indices                   | $N \times S$      |
| $\mathcal{Q}_G$          | Joint transition under policy $G$            | $M \times M$      |
| $\mathcal{R}_G^*, \pi$   | Chosen utility, joint invariant distribution | $M \times 1$ each |

Define  $c_{n,s,n'} = (1 + r^0)k_n + w^0 \bar{\epsilon}_s \bar{l} - k_{n'}$  and

$$\mathcal{R}_{m(n,s),n'} = \begin{cases} u(c_{n,s,n'}), & c_{n,s,n'} > 0, \\ -\infty, & c_{n,s,n'} \leq 0. \end{cases}$$

Since  $(\mathcal{P}\mathcal{V}^T)_{s,n'} = \sum_{s'} \mathcal{P}_{ss'} \mathcal{V}_{n',s'}$ , the stacked Bellman equation is

$$\boxed{\mathcal{V}(\cdot) = \mathcal{T}(x) = \max_{\text{columns}} \left[ \mathcal{R} + \beta(\mathcal{P}\mathcal{V}^T) \otimes \mathbf{1}_N \right]}. \quad (2)$$

The maximization is over savings choices. Its argmax gives  $\mathcal{G}(\cdot)$ , with the smallest index selected at an exact tie. Set  $g(k_n, s) = k_{\mathcal{G}(n,s)}$ . Then

$$\mathcal{Q}_G[m(n, s), m(n', s')] = \mathcal{P}_{ss'} \mathbf{1}\{n' = \mathcal{G}(n, s)\}, \quad \mathcal{R}_G^*[m(n, s)] = \mathcal{R}[m(n, s), \mathcal{G}(n, s)]. \quad (3)$$

Each row of  $\mathcal{Q}_G$  sums to one. For a fixed policy,  $x_G = \mathcal{R}_G^* + \beta \mathcal{Q}_G x_G$ ; the column distribution satisfies  $\mathcal{Q}_G^T \pi = \pi$ ,  $\mathbf{1}^T \pi = 1$ , and  $\pi \geq 0$ .

**3. Continuous-choice Euler equation.** Now allow continuous  $k' \geq \underline{k}$ . With multiplier  $\mu \geq 0$  on  $k' - \underline{k} \geq 0$ , write

$$\mathcal{L} = u((1 + r^0)k + w^0 \bar{\epsilon}_s \bar{l} - k') + \beta \sum_{s'} \mathcal{P}_{ss'} v(k', s') + \mu(k' - \underline{k}).$$

At an optimum, the first-order and complementary-slackness conditions are

$$-u'(c) + \beta \sum_{s'} \mathcal{P}_{ss'} v_k(k', s') + \mu = 0, \quad (4)$$

$$\mu \geq 0, \quad k' - \underline{k} \geq 0, \quad \mu(k' - \underline{k}) = 0. \quad (5)$$

The envelope condition, holding prices fixed, is

$$v_k(k, s) = (1 + r^0)u'(c(k, s)). \quad (6)$$

Evaluating the envelope condition at next period's state and using  $u'(c) = c^{-\sigma}$  gives

$$\boxed{c(k, s)^{-\sigma} = \beta(1 + r^0) \sum_{s'} \mathcal{P}_{ss'} c(k', s')^{-\sigma} + \mu.} \quad (7)$$

Thus current marginal utility is at least discounted expected next-period marginal utility times the gross return. If  $k' > \underline{k}$ , then  $\mu = 0$  and the Euler equation holds with equality. At the borrowing limit, it may be a strict inequality. This condition assumes no active artificial upper asset bound; choices at that grid boundary are flagged as a possible truncation issue.

**4. Euler residual for the discrete policy.** Using the grid policy, compute consumption by exact index lookup:

$$c_{n,s} = (1 + r^0)k_n + w^0 \bar{\epsilon}_s \bar{l} - k_{\mathcal{G}(n,s)}, \quad (8)$$

$$c_{n,s,s'}^{next} = (1 + r^0)k_{\mathcal{G}(n,s)} + w^0 \bar{\epsilon}_{s'} \bar{l} - k_{\mathcal{G}(\mathcal{G}(n,s),s')}. \quad (9)$$

Define the Euler-implied consumption and the absolute, unit-free consumption error as

$$c_{n,s}^{EE} = \left[ \beta(1 + r^0) \sum_{s'} \mathcal{P}_{ss'} (c_{n,s,s'}^{next})^{-\sigma} \right]^{-1/\sigma}, \quad \boxed{E_{n,s} = \left| 1 - \frac{c_{n,s}^{EE}}{c_{n,s}} \right|.} \quad (10)$$

For part (h), use all states in  $A = \{(n, s) : k_{\mathcal{G}(n,s)} > 0\}$ , flag upper-bound choices, and plot  $E_{n,s}$  against  $k_n$  for each shock. Report the maximum and arithmetic mean over  $A$ . With conditional weights  $q_{n,s} = \pi_{n,s} / \sum_A \pi_{n,s}$ , report  $\sum_A q_{n,s} E_{n,s}$  and  $\max_A (q_{n,s} E_{n,s})$ , labeling the latter the maximum weighted contribution. Also report the maximum over slack states with  $\pi_{n,s} > 10^{-12}$ . Empty sets or zero weight denominators give unavailable summaries. Discrete choices need not satisfy the continuous Euler equality exactly.

**5. Link to the M.2 kernel.** The kernel uses  $N = 100$  uniform points on  $[0, 20]$ , starts from  $\mathcal{V}_0 = 0$ , and stops when

$$\max_m \frac{|\mathcal{T}(x)_m - x_m|}{|x_m| + 1} \leq 10^{-10}.$$

It returns the current values and their greedy policy before another update; the iteration count includes the stopping check. Power iteration starts at  $\pi_0 = \mathbf{1}/M$  and returns the updated distribution once  $\|\pi_{j+1} - \pi_j\|_\infty \leq 10^{-12}$ . In Python, column stacking uses `order="F"`, and saved policy indices are zero-based. The script saves `V`, `G`, `pi`, and `iterations` in `kernel_output.npz`, matching the specification's interface.